

HUSENABASHA MAHAMMAD SOMPUR

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Summary

Electronics and Communication Engineering student with strong foundation in IoT, AI, and software development. Experienced in hardware-software integration projects using ESP32, Python programming, and machine learning applications. Demonstrated record of developing innovative solutions for real-world problems with focus on automation and intelligent systems. Proven expertise in embedded systems, networking, and AI model deployment with active GitHub portfolio showcasing technical projects.

Skills

- Programming Languages (Python, Java, C)
- Web Development (HTML, CSS, Flask)
- Data Structures & Algorithms
- Project Management
- Problem Solving
- Technical Documentation
- AI & Machine Learning (Python ML, NVIDIA DLI)
- IoT & Embedded (ESP32, Arduino, Sensors)
- Networking (Routing, QoS, Cisco GNS3)
- Development Tools (Git, MATLAB, Multisim)
- Hardware-Software Integration
- System Design & Analysis

Experience

AI Image Segmentation Developer

Personal Project | 2025

- Developed and deployed web application using Flask framework with AI models for image segmentation tasks.
- Achieved 92% segmentation accuracy using pre-trained deep learning models from NVIDIA DLI certification.
- Implemented secure file upload, image preprocessing, and real-time model inference capabilities.
- Successfully handled multiple simultaneous users with average processing time of 3 seconds per image.
- Created RESTful API for programmatic access and integrated user-friendly web interface.

Hardware Systems Developer

Personal Projects | 2024-2025

- Designed and implemented automatic headlight adjustment system using ESP32, achieving 95% accuracy.
- Developed gesture-controlled device with MPU6050 sensor, reaching 90% recognition accuracy.
- Built comprehensive attendance management system using PIR sensors and fingerprint authentication.
- Integrated multiple sensor technologies (LDR, Ultrasonic, PIR) for real-time environmental monitoring.
- Reduced manual intervention by 80% through innovative automation solutions.

Software Engineering Intern

TechnoHacks | May 2025 to June 2025

- Completed comprehensive Python development internship program covering advanced programming concepts.
- Developed load balancer implementation using Round Robin and Least Connection algorithms.
- Achieved 40% improvement in response time and handled 1000+ concurrent connections successfully.
- Created monitoring dashboard for real-time performance tracking and server status visualization.
- Gained expertise in socket programming, network communication, and system optimization.

Education and Training

Bachelor of Technology - Electronics and Communication Engineering

K L University | 3rd Year | Expected Graduation: 2026

- Prompt Engineering & Programming with OpenAI - Columbia University (2025)
- Python Development Internship - TechnoHacks (May-June 2025)
- AI for Business - Columbia University via Coursera (2025)
- Image Segmentation with AI - NVIDIA Deep Learning Institute (2025)
- Accenture Software Engineering Job Simulation - Forage (2025)

Languages

English:	Proficient	Telugu:	Native
Hindi:	Conversational	Technical Communication:	Advanced